

SMART UTM — a tensile rig that runs, watches and stops its own tests

Jönköping University · School of Engineering — Materials & Manufacturing

One specimen now yields a CURVE, not a point.

12 automation features and 6 closed-loop test protocols, every one validated on the rig. Camera-based strain sits INSIDE the control loop, so the machine measures the specimen instead of measuring itself.

12 / 16

smart features
built & validated

6

closed-loop test
protocols

17

runs auto-indexed,
13 specimens

$\pm 1.3\text{ N}$

force hold over
877 s

26 $\mu\epsilon$

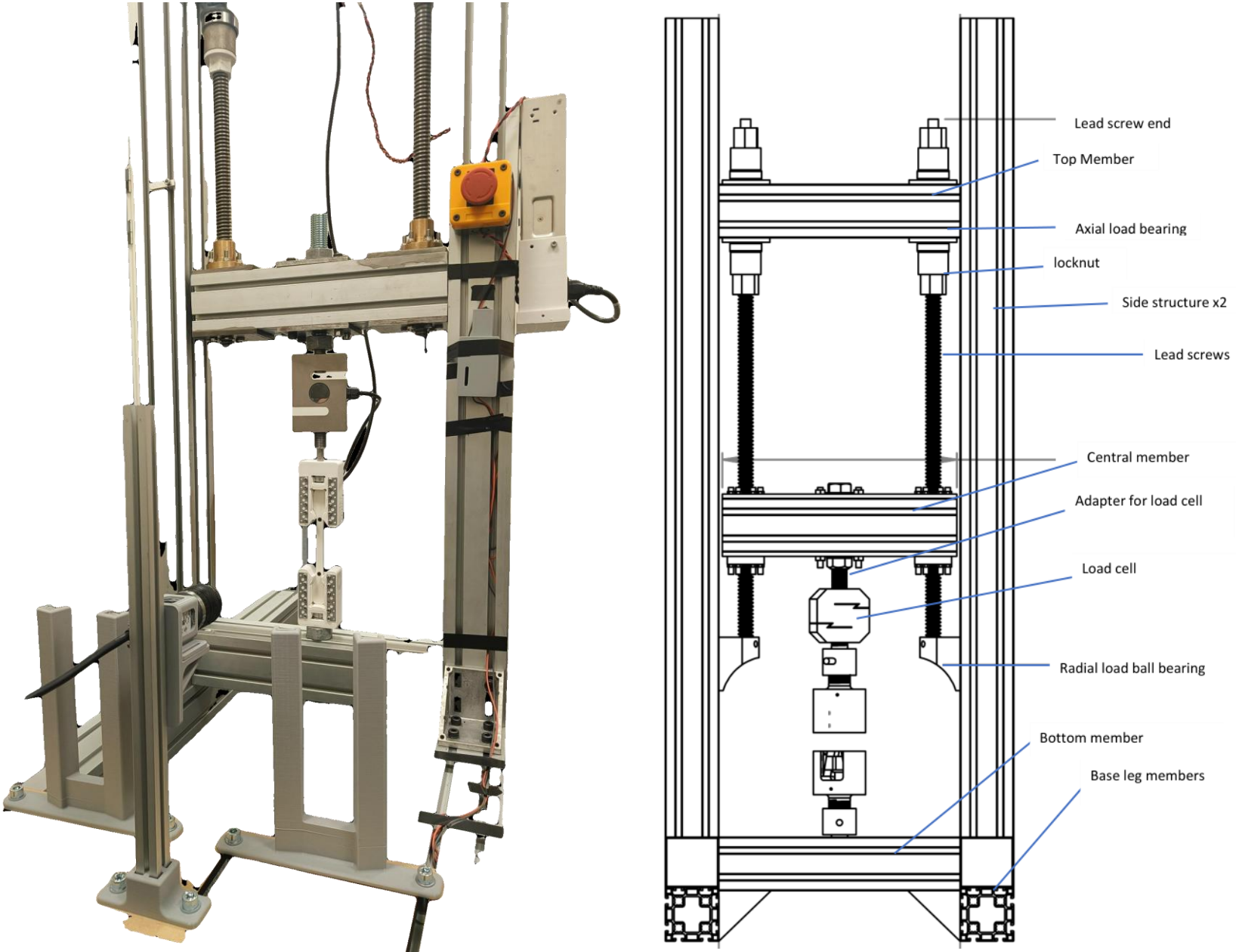
DIC noise floor
(900 s window)

0.9 %

agreement between two
fracture protocols

THE MACHINE

The rig — built in-house, every control loop closed in software



Lead screw end

Top Member

Axial load bearing

locknut

Side structure x2

Lead screws

Central member

Adapter for load cell

Load cell

Radial load ball bearing

Bottom member

Base leg members

Two steppers (1.85 Nm, 20:1) drive 5 mm lead screws against a fixed base. An ANYLOAD 3 t cell measures force; a Basler camera measures strain optically at 20.8 px/mm from two markers on the 80 mm gauge. The ESP32 runs STEP/DIR/ENABLE only and knows nothing of test protocols — every protocol, guard and measurement here lives in the PyQt6 app, closing the loop at ~11 Hz.

WHY AUTOMATE A TENSILE RIG

A conventional pull gives ONE modulus, ONE yield, ONE strength — and a human presses every button. Crosshead travel is not specimen strain, so the machine reports its own compliance as material behaviour. A destructive test is one irreversible shot. And nothing watches the run, so a stall, a lost marker or a fracture are all noticed late, by a person.

HOW IT WORKS

One control engine

All six protocols are one function called on every load sample (~11 Hz) with load, position and time-matched DIC strain in hand. A protocol only returns a speed — it cannot bypass the shared safety net.

Strain from the specimen

A camera tracks two markers; engineering strain is $(L-L_0)/L_0$ at 20.8 px/mm. The centroid resolves 0.1 px steps at ± 0.02 px.

Simulated before hardware

Every policy is replayed against a spring plant and recorded CSVs — 9/9 checks pass before a specimen is risked.

Four safety layers

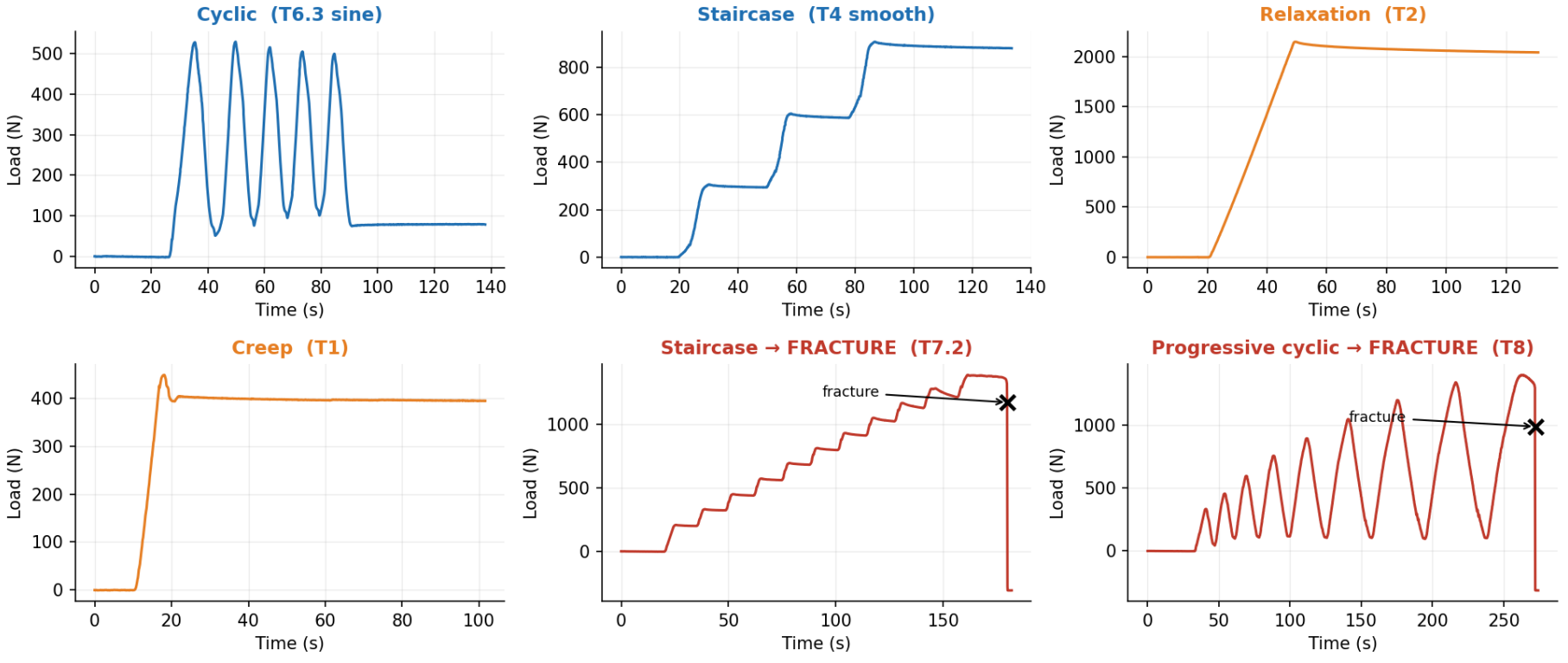
Load-collapse detector · stall guard ($< 0.05\text{ mm} / 6\text{ s}$ under load) · 10 kN and 30 mm backstops · dead-DIC guard (freeze at 0.2 s, halt at 1.0 s). Guards are phase-aware, so the stall guard stays silent through an intentional dwell.

One analysis library

The app, the reports and every figure compute E , σ_y , UTS, ϵ_f and the force anchor from the same code — a live readout and a published number cannot disagree.

WHAT THE AUTOMATION BUYS YOU

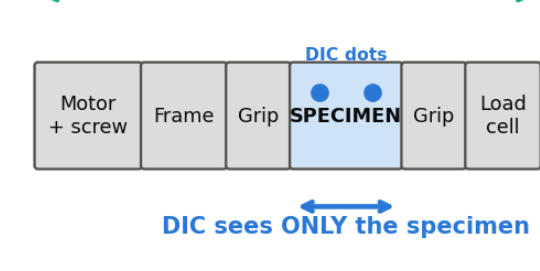
Six closed-loop protocols — every trace is real rig data



All six are one function behind one safety net; a protocol only chooses a speed. Two of them drive the specimen to destruction while interrogating it on the way — which is how one specimen yields a curve instead of a point. Two protocols taken independently to fracture landed 0.9 % apart on ultimate strength (21.19 vs 21.38 MPa) — on different specimens, $n = 1$ each, so that bounds protocol and specimen scatter together.

The machine reads STIFFER while the specimen SOFTENS

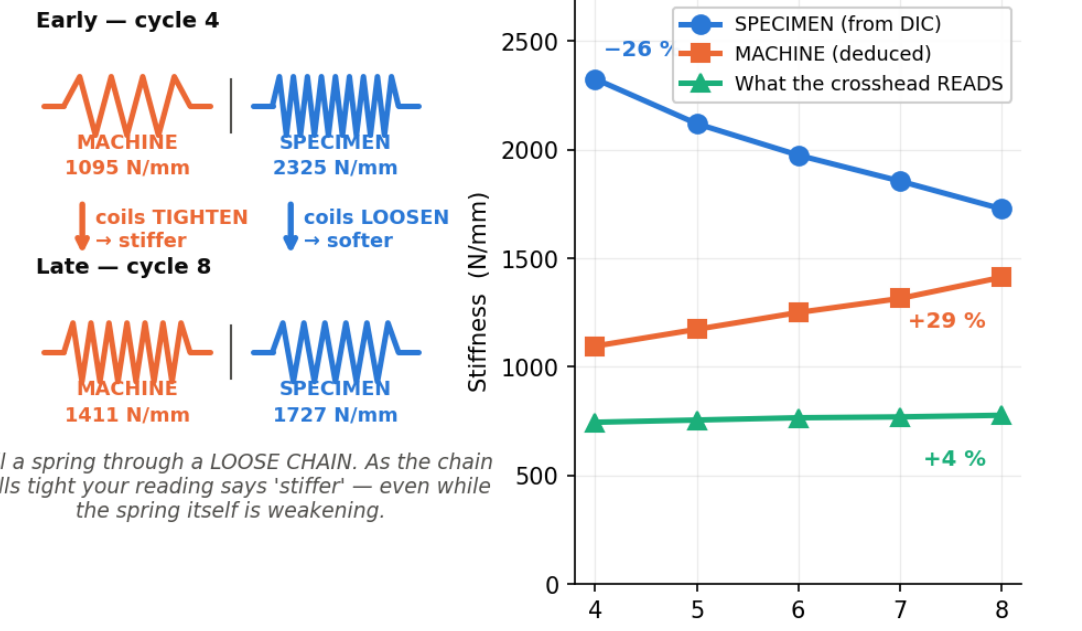
1 · The two rulers are not the same length



CROSSHEAD sees ALL of this

DIC sees ONLY the specimen

2 · They are two springs in a row



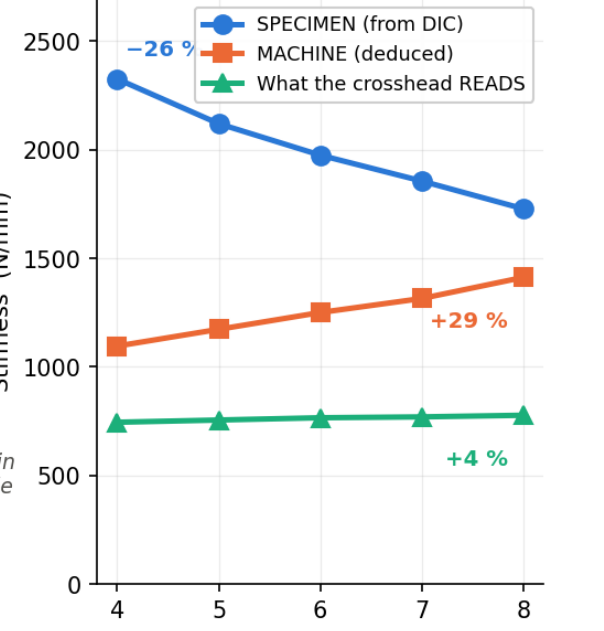
Early — cycle 4

Late — cycle 8

Machine stiffness: 1095 N/mm → 1411 N/mm (+29%)

Specimen stiffness: 2325 N/mm → 1727 N/mm (-26%)

3 · Same units, one axis — now it is obvious

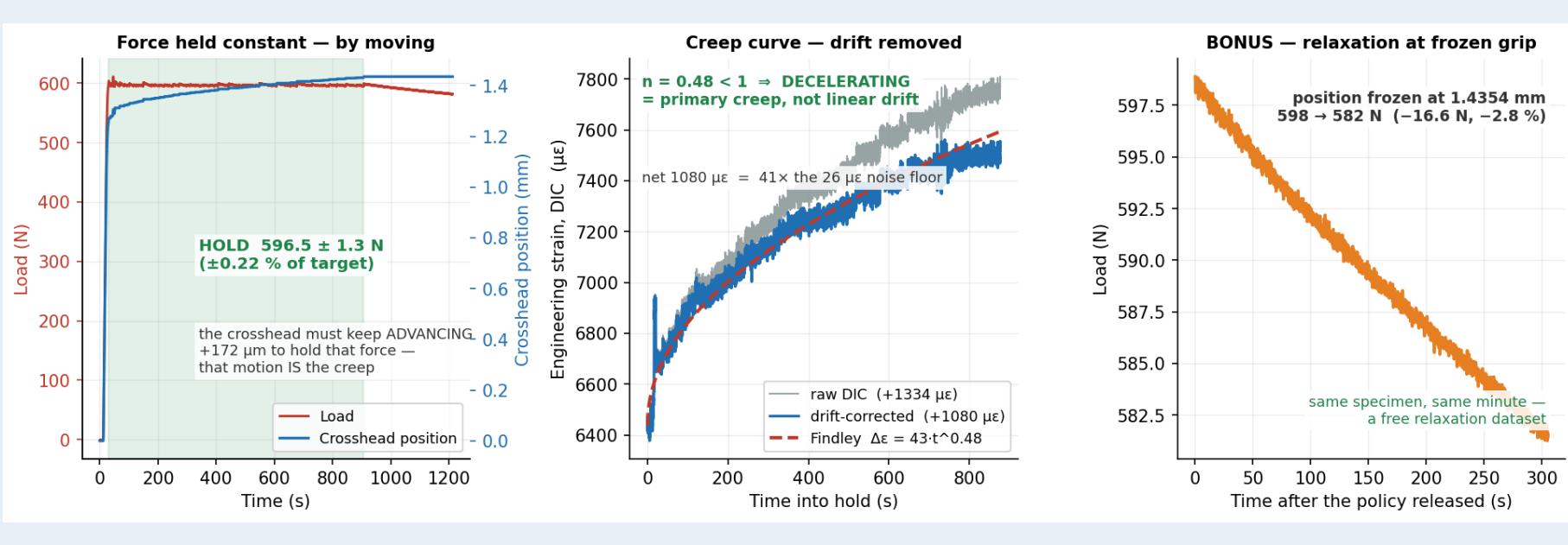


Stiffness (N/mm)

Cycle

Progressive cyclic to fracture. Crosshead stiffness ROSE 744 → 777 N/mm as rig slack squeezed out, while the DIC-measured specimen modulus FELL 2.32 → 1.73 GPa, a 26 % loss. Read from the crosshead alone this test says the material is getting stiffer — the strongest argument for putting the camera in the loop.

Creep, resolved — and separated from the instrument's own drift



Two holds, in order. A 928 s ZERO-LOAD baseline measured the rig's own drift at $+0.2893\text{ }\mu\epsilon/\text{s}$; the real hold then pinned $597 \pm 1.3\text{ N}$ for 877 s at 11.3 MPa. Raw strain minus that slope leaves $+1080\text{ }\mu\epsilon$ of creep — $41\times$ the $26\text{ }\mu\epsilon$ noise floor — and it decelerates ($n = 0.48$), the criterion fixed BEFORE the run for telling creep from drift. Holding it took $+172\text{ }\mu\text{m}$ of crosshead but only $+107\text{ }\mu\text{m}$ of gauge: 62 % specimen.

LIMITATIONS

Poisson's ratio is not measurable today

The gauge is too narrow: the elastic width change is sub-pixel at 20.8 px/mm. Every stress here is ENGINEERING (F/A_0), never 'true'.

No fatigue damage curve yet

One run lost the camera mid-way and its repeat started on an already-damaged specimen, and the modulus RECOVERED between them — so no valid E_0 exists.

Creep and fatigue are each $n = 1$

Both measurements are proven; the material constants are not. Creep needs 3–4 stress levels and fatigue needs repeats.

The motor's torque ceiling is variable

Normally 3.2–3.4 kN, but ~2.6 kN on one session. The drivetrain is rated far above that, so the cause is electrical — not the software.

THE SMART FEATURES, SF1 TO SF16

built & rig-validated (12)

Live DIC health HUD

A green/amber/red badge shows marker lock, tracking % and jitter live — so a dropout is seen during the test, not found later in the CSV.

SF1

2 / 2

markers, live at ~2 Hz

planned (3)

Prepare specimen

One button tares position, force and DIC and clears the plots. It tares DIC only at a green 2/2 lock, so a bad mount cannot be zeroed in.

SF2

4 → 1

operator actions

hardware-blocked (1)

Recipes / settings

Saves and restores an entire setup, including all six protocols' parameters — so repeats are identical by construction, not by memory.

SF3

1 click

restores the whole setup

One-click report

A one-page PDF and every plot, recomputed from the CSV by the same library the app runs live.

SF4

1 click

→ PDF + every graph

Stall guard

Halts a motor that has stopped moving under load. It proved the T7 failure was a 2.6 kN stall, not a fracture — the specimen came off intact.

SF7

0.05 mm / 6 s

trip, armed above 200 N

Auto-preload

Approaches on a 0.2 → 0.1 → 0.02 mm/s schedule and stops 3 % high, because PLA relaxes about 2 % while held.

SF10

1.03 ×

target, pre-paying relaxation

Guided workflow + live overlay

A Connect→Calibrate→Mount→Prepare→Run→Save wizard, with live modulus and a fracture flag as the test runs.

SF13

planned

usability

Dead-DIC guard + hard backstops

A controller ramping blind on a stale camera reading spikes the force. Speed freezes at 0.2 s of frozen strain, halt at 1.0 s.

SF16

518 → 175 N

overshoot on marker loss

Auto-stop at fracture

A load-collapse detector halts the motor the moment the specimen breaks. Caught S16 at 3375 N, silent through the ductile draw before it.

SF5

1.2 s

collapse → motor halt

Release load

Taring at preload makes the readout say 0 N while 300 N is still on the specimen. This drives past that, down to the true zero.

SF8

≤ 5 N

residual, at 0.20 mm/s

Auto-metadata + foldering

Per-specimen folders and file IDs filled from the recipe, plus a one-click per-specimen slide deck.

SF11

planned

next feature up

Poisson's ratio / true stress

The four-marker maths is written and self-tested. The gauge is too narrow for a transverse pair at 20.8 px/mm.

SF14

blocked

optics, not code

Constant-strain-rate testing

Closed loop on the specimen's own $d\epsilon/dt$: the crosshead adapted itself 0.10 → 0.05 mm/s, fast in the elastic region, slow through necking.

SF6

0.00051 / s

held, vs a 0.0005 target

Six closed-loop protocols

cyclic · staircase · relaxation · creep · staircase→FRACTURE · progressive-cyclic→FRACTURE. One engine, one safety net.

SF9

6 / 6

rig-validated

DIC auto-calibrate

Exposure and threshold swept on camera start, and an ROI that follows the markers through a ductile draw.

SF12

planned

camera robustness

Test registry

Each run indexed with whatever its loading path supports — UTS, ϵ_f , E , σ_y , toughness — plus the force anchor that turns a tared reading into a true load.

SF15

17 runs

auto-indexed, 13 specimens

Code, raw CSVs and the full 76-slide technical deck: [Software/UTM_PyQt6/](#) and [documentation/](#) · Every figure and number here is regenerated from the rig CSVs by [documentation/sf9_data.py](#) — none of it is transcribed by hand. · Layout follows the [#evenbetterposter](#) / [Generation-2 billboard](#) format: a plain-language headline, with methods and limitations kept as full panels rather than sidebars.